

HDOC 8

Question 1:

Write a C program to find the remaining volume of a cylinder after removing a cone of the same radius and height from it.

Understanding the Problem Statement

1. Input

The program takes:

- Radius of the cone/cylinder
- Height of the cylinder/height

Required Output

The program should calculate:

- Vol of cylinder - Vol of cone

Formula Used:

$$\text{Vol} = \pi r^2 h - (\pi r^2 h) / 3 = \frac{2}{3} (\pi r^2 h)$$

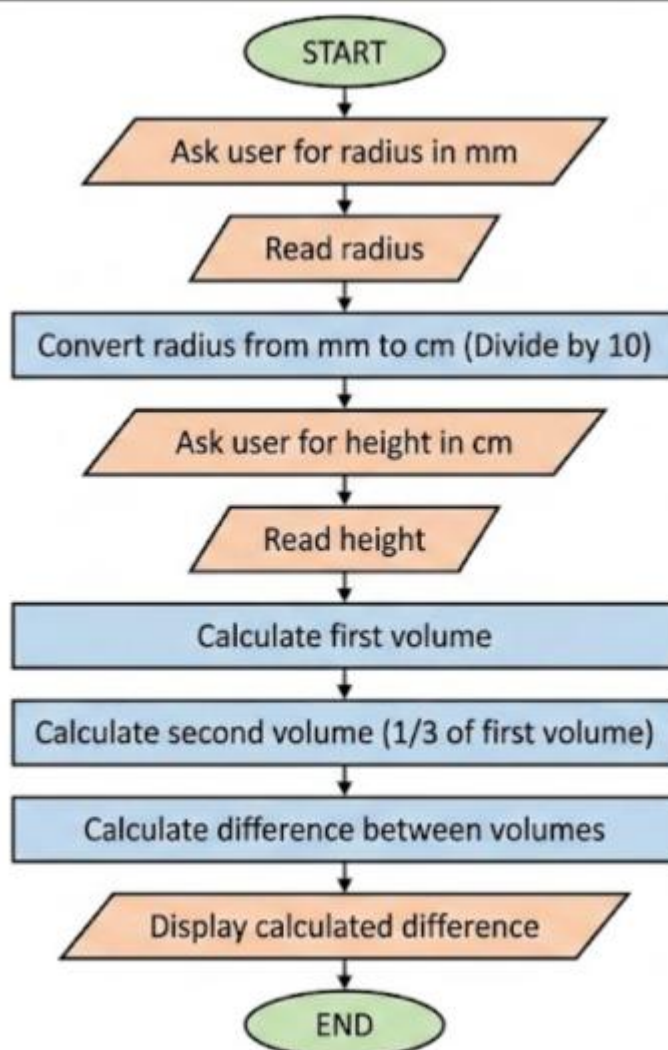
2. Standard Test Case Table

Test Case	Input	Expected Output	Actual Output	Result
1	R=70,h=10	Vol= 1026.252	Vol=1026.252808	Pass
2	R=100,h=15	Vol=3141.589	Vol=3141.589844	Pass
3	R=120,h=21	Vol=6333.445	Vol=6333.445312	Pass
4	R=150,h=25	Vol=11780.961	Vol=11780.961914	Pass
5	R=170,h=30	Vol= 18158.390	Vol= 18158.390625	Pass

3. Evaluation of the Algorithm

The algorithm takes the input for the value of radius and height of cylinder, then it calculates the volume of the cylinder and subsequently it calculates the volume of the cone with the same measurements(given). Then to get the desired output, the algorithm gives out the difference between the two volumes, and prints the output using printf function.

4. FLOWCHART



5. C- Program

```
#include<stdio.h>
```

```
#define PI 3.14159
```

```
int main()
```

```
{
```

```
    float radius, height;
```

```

printf("Enter the value of the radius in mm");

scanf("%f",&radiusm);

float radiusc=radiusm/10;

printf("Enter the height in cm");

scanf("%f",&height);

float vol1=PI*radiusc*radiusc*height;

float vol2= (PI*radiusc*radiusc*height)/3;

float op=vol1-vol2;

printf("The difference between the
vol is: %f cm3",op);

return 0;

}

```

Save the program with the filename:

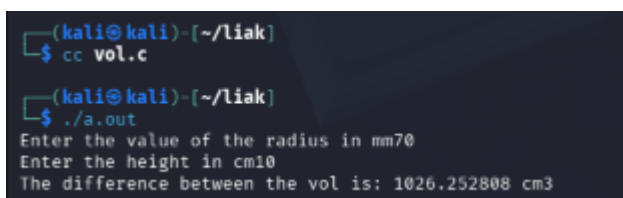
vol.c

Compile it using CC:

cc vol.c

If there are no errors, execute the program using:

./a.out



```

(kali@kali) - [~/liak]
$ cc vol.c

(kali@kali) - [~/liak]
$ ./a.out
Enter the value of the radius in mm: 70
Enter the height in cm: 10
The difference between the vol is: 1026.252808 cm3

```

6. Sample Execution:

Test Case:

Enter the value of the radius in mm: **70**

Enter the height in cm: **10**

The difference between the vol is : **1026.252808 cm3**

7. Evaluation against test cases:

The program was tested using different value of radius and height including large and small values. In each case the difference in the volume was coming to be correct when verified manually.

Therefore, All the test cases were successfully passed.

8. RESULT

Thus, the C program to calculate the Difference between the volume of cone and cylinder with same height and radius when radius is in mm and height in cm was successfully written, compiled and executed. The program was tested with multiple test cases and produced expected results.

```
(kali㉿kali)-[~/liak]
$ cc vol.c

(kali㉿kali)-[~/liak]
$ ./a.out
Enter the value of the radius in mm70
Enter the height in cm10
The difference between the vol is: 1026.252808 cm3
```

Question 2:

Write a C program to find the area of a rhombus using side and included angle.

Understanding the Problem Statement

1. Input

The program calculates the area of a rhombus using its sides and the angle.

Required Output

The program should calculate:

- Area of the rhombus.

Formula Used:

S= Side of the rhombus

Area= $s^2 \cdot \sin \theta$

2. Standard Test Case Table

Test Case	Side	Expected Output	Actual Output	Result
1	S=10 Theta= 45	Area=70.710	Area=70.710	Pass
2	S=15 Theta=65	Area=203.919	Area=203.919	Pass
3	S=20 Theta=75	Area=386.370	Area=386.370	Pass
4	S=30 Theta=50	Area= 689.389	Area= 689.389	Pass
5	S=25 Theta=35	Area= 358.484	Area= 358.484	Pass

3. Evaluation of the Algorithm

The algorithm helps us find the area of cross section of the wire using the formula

$\text{Area} = \text{Side}^2 (\sin(x))$, The program takes the value of side and angle between two sides. Then the program works in such a way that it finds out the area of the rhombus

4. C- Program

```
#include <stdio.h>
```

```
#include <math.h>
```

```
int main()
```

```
{
```

```
    float side, angle, area;
```

```
    printf("Enter the side of the rhombus: ");
```

```
    scanf("%f", &side);
```

```
    printf("Enter the angle between the sides  
in degrees: ");
```

```
    scanf("%f", &angle);
```

```
    angle = angle * 3.14159 / 180;
```

```
    area = side * side * sin(angle);
```

```
    printf("Area of the rhombus is %f", area);
```

```
    return 0;
```

```
}
```

Save the program with the filename:

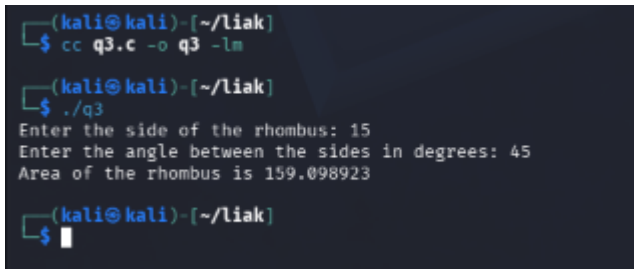
gedit q3.c

Compile it using CC:

cc q3.c -o q3 -lm

If there are no errors, execute the program using:

./q3



```
(kali@kali)~/.liak
$ cc q3.c -o q3 -lm
(kali@kali)~/.liak
$ ./q3
Enter the side of the rhombus: 15
Enter the angle between the sides in degrees: 45
Area of the rhombus is 159.098923
(kali@kali)~/.liak
$
```

5. Sample Execution:

Test Case:

Enter the side of the rhombus: 15

Enter the angle between the sides in degrees: 45

Area of the rhombus is 159.098923

6. Evaluation against test cases:

The program was tested using different value of side of rhombus and angle between two sides including large and small values. In each case the Area of rhombus was coming to be correct when verified manually.

Therefore, All the test cases were successfully passed.

7. RESULT

Thus, the C program to calculate the Area of rhombus when the side of the rhombus and the angle between the two sides were given was successfully written, compiled and executed. The program was tested with multiple test cases and produced expected results.

```
(kali㉿kali)-[~/liak]
$ cc q3.c -o q3 -lm

(kali㉿kali)-[~/liak]
$ ./q3
Enter the side of the rhombus: 15
Enter the angle between the sides in degrees: 45
Area of the rhombus is 159.098923

(kali㉿kali)-[~/liak]
$ ./q3
Enter the side of the rhombus: 10
Enter the angle between the sides in degrees: 45
Area of the rhombus is 70.710632

(kali㉿kali)-[~/liak]
$ ss
```

Name- Keshav Mittal

Batch -16

SAP ID- 590032934